

VECTOR 3D

Vector 3D

1. If $\vec{a}$ is a nonzero vector such that its projections on the vectors $2\hat{i} - \hat{j} + 2\hat{k}$, $\hat{i} + 2\hat{j} - 2\hat{k}$ and $\hat{k}$ are equal, then a unit vector along $\vec{a}$ is :

(1) $\frac{1}{\sqrt{155}}(-7\hat{i} + 9\hat{j} + 5\hat{k})$

(2) $\frac{1}{\sqrt{155}}(7\hat{i} + 9\hat{j} + 5\hat{k})$

(3) $\frac{1}{\sqrt{155}}(-7\hat{i} + 9\hat{j} - 5\hat{k})$

(4) $\frac{1}{\sqrt{155}}(7\hat{i} + 9\hat{j} - 5\hat{k})$

Ans. B

Sol. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$$a_1^2 + a_2^2 + a_3^2 = 1$$

Let $\vec{b} = 2\hat{i} - \hat{j} + 2\hat{k}$, $\vec{c} = \hat{i} - 2\hat{j} - 2\hat{k}$

$$\vec{d} = \hat{k}$$

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{\vec{a} \cdot \vec{c}}{|\vec{c}|} = \frac{\vec{a} \cdot \vec{d}}{|\vec{d}|}$$

$$\frac{2a_1 - a_2 + 2a_3}{3} = \frac{a_1 + 2a_2 - 2a_3}{3} = a_3$$

By solving

$$a_1 = \frac{7}{\sqrt{155}}, a_2 = \frac{9}{\sqrt{155}}, a_3 = \frac{5}{\sqrt{155}}$$

2. Let $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} + 2\hat{j} + 5\hat{k}$ and a vector $\vec{c}$ be such that $(\vec{a} - \vec{c}) \times \vec{b} = -18\hat{i} - 3\hat{j} + 12\hat{k}$ and $\vec{a} \cdot \vec{c} = 3$. If $\vec{b} \times \vec{c} = \vec{d}$, then $|\vec{a} \cdot \vec{d}|$ is equal to :

माना $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} + 2\hat{j} + 5\hat{k}$ और एक सदिश $\vec{c}$ के लिए $(\vec{a} - \vec{c}) \times \vec{b} = -18\hat{i} - 3\hat{j} + 12\hat{k}$ तथा $\vec{a} \cdot \vec{c} = 3$ हैं। यदि

$\vec{b} \times \vec{c} = \vec{d}$ है, तो $|\vec{a} \cdot \vec{d}|$ बराबर है :

(1) 12

(2) 18

(3) 15

(4) 9

Ans. C

Sol. $(\vec{a} - \vec{c}) \times \vec{b} = -18\hat{i} - 3\hat{j} + 12\hat{k}$

$$\vec{a} \times \vec{b} - \vec{c} \times \vec{b} = -18\hat{i} - 3\hat{j} + 12\hat{k}$$

$$\Rightarrow \vec{a} \times \vec{b} + \vec{d} = -18\hat{i} - 3\hat{j} + 12\hat{k}$$

Take dot with $\vec{a}$

$$\vec{a} \cdot \vec{d} = -36 + 9 + 12 = -15$$

$$\therefore |\vec{a} \cdot \vec{d}| = 15$$

3. The line L_1 is parallel to vector $\vec{a} = -3\hat{i} + 2\hat{j} + 4\hat{k}$ and passes through the point $(7, 6, 2)$ and the line L_2 is parallel to the vector $\vec{b} = 2\hat{i} + \hat{j} + 3\hat{k}$ and passes through the point $(5, 3, 4)$. The shortest distance between the lines L_1 and L_2 is :

रेखा L_1 सदिश $\vec{a} = -3\hat{i} + 2\hat{j} + 4\hat{k}$ के समान्तर है और बिन्दु $(7, 6, 2)$ से होकर जाती है तथा रेखा L_2 सदिश $\vec{b} = 2\hat{i} + \hat{j} + 3\hat{k}$ के समान्तर है और बिन्दु $(5, 3, 4)$ से होकर जाती है। रेखाओं L_1 और L_2 के बीच न्यूनतम दूरी है :

- (1) $\frac{23}{\sqrt{38}}$ (2) $\frac{21}{\sqrt{57}}$ (3) $\frac{23}{\sqrt{57}}$ (4) $\frac{21}{\sqrt{38}}$

Ans. A

Sol. Equation of L_1 : $7\hat{i} + 6\hat{j} + 2\hat{k} + \lambda(-3\hat{i} + 2\hat{j} + 4\hat{k})$

Equation of L_2 : $5\hat{i} + 3\hat{j} + 4\hat{k} + \lambda(2\hat{i} + \hat{j} + 3\hat{k})$

Shortest distance between L_1 and L_2

$$\begin{aligned} &= \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|} \\ &= \frac{|(-2\hat{i} - 3\hat{j} + 2\hat{k}) \cdot (2\hat{i} + 17\hat{j} - 7\hat{k})|}{\sqrt{2^2 + (-17)^2 + (-7)^2}} \\ &= \frac{|-4 - 51 - 14|}{\sqrt{4 + 289 + 49}} \\ &= \frac{|-69|}{\sqrt{342}} = \frac{23}{\sqrt{38}} \end{aligned}$$

4. Consider two vectors $\vec{u} = 3\hat{i} - \hat{j}$ and $\vec{v} = 2\hat{i} + \hat{j} - \lambda\hat{k}, \lambda > 0$. The angle between them is given by $\cos^{-1}\left(\frac{\sqrt{5}}{2\sqrt{7}}\right)$.

Let $\vec{v} = \vec{v}_1 + \vec{v}_2$, where $\vec{v}_1$ is parallel to $\vec{u}$ and $\vec{v}_2$ is perpendicular to $\vec{u}$. Then the value $|\vec{v}_1|^2 + |\vec{v}_2|^2$ is equal to

:

दो सदिशों $\vec{u} = 3\hat{i} - \hat{j}$ और $\vec{v} = 2\hat{i} + \hat{j} - \lambda\hat{k}, \lambda > 0$ का विचार कीजिए। इनके बीच का कोण $\cos^{-1}\left(\frac{\sqrt{5}}{2\sqrt{7}}\right)$ है।

माना $\vec{v} = \vec{v}_1 + \vec{v}_2$ है, जहाँ $\vec{v}_1$ सदिश $\vec{u}$ के समान्तर है और $\vec{v}_2$ सदिश $\vec{u}$ के लंबवत है। तो $|\vec{v}_1|^2 + |\vec{v}_2|^2$ का मान है :

- (1) 14 (2) $\frac{25}{2}$ (3) $\frac{23}{2}$ (4) 10

Ans. A

Sol. $\vec{u} = 3\hat{i} - \hat{j}$

$$\vec{v} = 2\hat{i} + \hat{j} - \lambda\hat{k}; \lambda > 0$$

$$\cos \theta = \frac{6-1}{\sqrt{10}\sqrt{5+\lambda^2}} = \frac{\sqrt{5}}{2\sqrt{7}}$$

$$\frac{\sqrt{5}}{\sqrt{10}\sqrt{5+\lambda^2}} = \frac{\sqrt{5}}{\sqrt{2}\sqrt{7}}$$

$$5 + \lambda^2 = 14$$

$$\lambda^2 = 14 - 5$$

$$\lambda = \sqrt{9}$$

$$\lambda = 3$$

$$\therefore \vec{v} = 2\hat{i} + \hat{j} - 3\hat{k}$$

$$\vec{v} = \vec{v}_1 + \vec{v}_2$$

$$= \alpha(3\hat{i} - \hat{j}) + \vec{v}_2$$

$$\Rightarrow \vec{v}_2 = \vec{v} - \alpha(3\hat{i} - \hat{j})$$

$$= 2\hat{i} + \hat{j} - 3\hat{k} - 3\alpha\hat{i} + \alpha\hat{j}$$

$$\vec{v}_2 = (2-3\alpha)\hat{i} + (1+\alpha)\hat{j} - 3\hat{k}$$

Now, $\vec{v}_2$ is $\perp$ to $\vec{u}$

$$\vec{v}_2 \cdot \vec{u} = 0$$

$$(2-3\alpha)3 + (1+\alpha)(-1) = 0$$

$$6 - 9\alpha - 1 - \alpha = 0$$

$$5 = 10\alpha$$

$$\alpha = \frac{1}{2}$$

$$\left\{ \begin{array}{l} \vec{v}_1 \parallel \vec{u} \\ \therefore \vec{v}_1 = \lambda \vec{u} \end{array} \right\}$$

$$\vec{v}_1 = \frac{3\hat{i}}{2} - \frac{\hat{j}}{2}$$

$$|\vec{v}_1|^2 = \frac{9}{4} + \frac{1}{4} = \frac{10}{4}$$

$$= \frac{5}{2}$$

$$\vec{v}_2 = \left(2 - \frac{3}{2}\right)\hat{i} + \left(1 + \frac{1}{2}\right)\hat{j} - 3\hat{k}$$

$$= \frac{1}{2}\hat{i} + \frac{3}{2}\hat{j} - 3\hat{k}$$

$$|\vec{v}_2|^2 = \frac{1}{4} + \frac{9}{4} + 9$$

$$\Rightarrow \frac{10}{4} + 9$$

$$\Rightarrow \frac{5}{2} + 9$$

$$|\vec{v}_1|^2 + |\vec{v}_2|^2 = 5 + 9 = 14$$

5. Let the angle $\theta, 0 < \theta < \frac{\pi}{2}$ between two unit vectors $\hat{a}$ and $\hat{b}$ be $\sin^{-1}\left(\frac{\sqrt{65}}{9}\right)$. If the vector $\vec{c} = 3\hat{a} + 6\hat{b} + 9(\hat{a} \times \hat{b})$,

then the value of $9(\vec{c} \cdot \hat{a}) - 3(\vec{c} \cdot \hat{b})$ is :

माना दो मात्रक सदिशों $\hat{a}$ और $\hat{b}$ के बीच कोण $\theta = \sin^{-1}\left(\frac{\sqrt{65}}{9}\right), 0 < \theta < \frac{\pi}{2}$ है। यदि सदिश $\vec{c} = 3\hat{a} + 6\hat{b} + 9(\hat{a} \times \hat{b})$ है, तो

$9(\vec{c} \cdot \hat{a}) - 3(\vec{c} \cdot \hat{b})$ का मान है :

(1) 29

(2) 27

(3) 31

(4) 24

Ans. A

Sol. $\vec{c} = 3\vec{a} + 6\vec{b} + 9(\vec{a} \times \vec{b})$

$$\sin^{-1}\left(\frac{\sqrt{65}}{9}\right) \Rightarrow \sin \theta = \frac{\sqrt{65}}{9} \Rightarrow \cos \theta = \frac{4}{9}$$

$$\vec{c} \cdot \vec{a} = 3|\vec{a}|^2 + 6\vec{a} \cdot \vec{b} = 3 + \frac{6 \cdot 4}{9} = \frac{51}{9}$$

$$\vec{c} \cdot \vec{a} = 3\vec{a} \cdot \vec{b} + 6|\vec{b}|^2 = \frac{3 \cdot 4}{9} + 6 = \frac{22}{3}$$

$$\therefore 9(\vec{c} \cdot \vec{a}) - 3(\vec{c} \cdot \vec{b}) = 51 - 22 = 29$$

6. If the equation of the line passing through the point $\left(0, -\frac{1}{2}, 0\right)$ and perpendicular to the lines $\vec{r} = \lambda(\hat{i} + \hat{a}\hat{j} + b\hat{k})$

and $\vec{r} = (\hat{i} - \hat{j} - 6\hat{k}) + \mu(-b\hat{i} + \hat{a}\hat{j} + 5\hat{k})$ is $\frac{x-1}{-2} = \frac{y+4}{d} = \frac{z-c}{-4}$, then $a + b + c + d$ is equal to :

यदि बिन्दु $\left(0, -\frac{1}{2}, 0\right)$ से होकर जाने वाली तथा रेखाओं $\vec{r} = \lambda(\hat{i} + \hat{a}\hat{j} + b\hat{k})$ और $\vec{r} = (\hat{i} - \hat{j} - 6\hat{k}) + \mu(-b\hat{i} + \hat{a}\hat{j} + 5\hat{k})$ के

लम्बवत् रेखा का समीकरण $\frac{x-1}{-2} = \frac{y+4}{d} = \frac{z-c}{-4}$ है, तो $a + b + c + d$ बराबर है :

(1) 14

(2) 10

(3) 13

(4) 12

Ans. A

Sol. $\frac{x-1}{-2} = \frac{y+4}{d} = \frac{z-c}{-4}$

$\left(0, -\frac{1}{2}, 0\right)$ will satisfy it

$$\frac{1}{2} = \frac{7}{2d} = \frac{c}{4}$$

$$\Rightarrow d = 7, c = 2$$

Line is perpendicular to other lines

So,

$$(-2, d, -4) \cdot (1, a, b) = 0$$

$$\Rightarrow -2 + 7a - 4b = 0 \quad \dots(1)$$

and

$$(-2, d, -4) \cdot (-b, a, 5) = 0$$

$$\Rightarrow 2b + 7a - 20 = 0 \quad \dots(2)$$

from (1) and (2)

$$a = 2, b = 3$$

$$\text{so, } a + b + c + d = 2 + 3 + 2 + 7 = 14$$

7. Let $\vec{a}$ and $\vec{b}$ be the vectors of the same magnitude such that $\frac{|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}|}{|\vec{a} + \vec{b}| - |\vec{a} - \vec{b}|} = \sqrt{2} + 1$. Then $\frac{|\vec{a} + \vec{b}|^2}{|\vec{a}|^2}$ is :

माना बराबर परिमाण के दो सदिश $\vec{a}$ और $\vec{b}$ इस प्रकार हैं कि $\frac{|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}|}{|\vec{a} + \vec{b}| - |\vec{a} - \vec{b}|} = \sqrt{2} + 1$ है। तो $\frac{|\vec{a} + \vec{b}|^2}{|\vec{a}|^2}$ बराबर है :

(1) $2 + 4\sqrt{2}$

(2) $1 + \sqrt{2}$

(3) $2 + \sqrt{2}$

(4) $4 + 2\sqrt{2}$

Ans. C

Sol. Let $|\vec{a}| = |\vec{b}| = P$

and $\vec{a} \wedge \vec{b} = \theta$

$$|\vec{a} + \vec{b}| = \sqrt{2}P\sqrt{1 + \cos \theta} = P \cdot 2 \cos \frac{\theta}{2}$$

$$|\vec{a} - \vec{b}| = P \cdot 2 \sin \frac{\theta}{2}$$

$$\text{So, } \frac{|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}|}{|\vec{a} + \vec{b}| - |\vec{a} - \vec{b}|} = \frac{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} - \sin \frac{\theta}{2}} = \frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}}$$

$$\Rightarrow \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) = \sqrt{2} + 1$$

$$\Rightarrow \frac{\pi}{4} + \frac{\theta}{2} = \frac{3\pi}{8} \Rightarrow \theta = \frac{\pi}{4}$$

$$\text{Now, } \frac{|\vec{a} + \vec{b}|^2}{|\vec{a}|^2} = \frac{4P^2 \cos^2 \frac{\theta}{2}}{P^2}$$

$$= 4 \cos^2 \frac{\pi}{8}$$

$$= 2 \left(1 + \cos \frac{\pi}{4} \right) = 2 \left(1 + \frac{1}{\sqrt{2}} \right)$$

$$= \sqrt{2} (\sqrt{2} + 1)$$

$$= 2 + \sqrt{2}$$

8. Let the vertices Q and R of the triangle PQR lie on the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$, QR = 5 and the coordinates of the point P be (0, 2, 3). If the area of the triangle PQR is $\frac{m}{n}$:

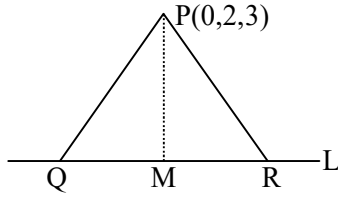
माना त्रिभुज PQR के शीर्ष Q और R रेखा $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$ पर हैं, QR = 5 है और बिन्दु P के निर्देशांक (0, 2, 3) हैं। यदि

त्रिभुज PQR का क्षेत्रफल $\frac{m}{n}$ है, तो :

- (1) $2m - 5\sqrt{21}n = 0$ (2) $m - 5\sqrt{21}n = 0$ (3) $5m - 2\sqrt{21}n = 0$ (4) $5m - 21\sqrt{2}n = 0$

Ans. A

Sol.



$$QR = 5$$

$$M(5\lambda - 3, 2\lambda + 1, 3\lambda - 4)$$

$$\text{Drs of PM} \Rightarrow 5\lambda - 3, 2\lambda - 1, 3\lambda - 7$$

$$\text{Drs of line L} \Rightarrow 5, 2, 3$$

$$PM \perp L$$

$$\Rightarrow (5\lambda - 3)5 + (2\lambda - 1)2 + (3\lambda - 7)3 = 0$$

$$\Rightarrow \lambda = 1$$

$$\therefore M(2, 3, -1)$$

$$PM = \sqrt{4+1+16} = \sqrt{21}$$

$$\text{Area} = \frac{1}{2} \times 5 \times \sqrt{21} = \frac{m}{n}$$

$$2m - 5\sqrt{21}n = 10$$

9. Let ABCD be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the areas of the triangles ABC, ACD and ADB be 5, 6 and 7 square units respectively. Then the area (in square units) of the ΔBCD is equal to :

माना ABCD एक चतुष्फलक है जिसके किनारे AB, AC और AD परस्पर लंबवत हैं। माना त्रिभुजों ABC, ACD और ADB के क्षेत्रफल क्रमशः 5, 6 और 7 वर्ग इकाई हैं। तो त्रिभुज BCD का क्षेत्रफल (वर्ग इकाई में) बराबर है :

- (1) $7\sqrt{3}$ (2) $\sqrt{110}$ (3) $\sqrt{340}$ (4) 12

Ans. B

Sol. $\text{Ar}(\Delta BCD)$

$$= \sqrt{(\text{Ar}(\Delta ABC))^2 + (\text{Ar}(\Delta ACD))^2 + (\text{Ar}(\Delta ADB))^2}$$

$$= \sqrt{5^2 + 6^2 + 7^2}$$

$$= \sqrt{110}$$

10. Line L_1 passes through the point $(1, 2, 3)$ and is parallel to z -axis. Line L_2 passes through the point $(\lambda, 5, 6)$ and is parallel to y -axis. Let for $\lambda = \lambda_1, \lambda_2, \lambda_2 < \lambda_1$, the shortest distance between the two lines be 3. Then the square of the distance of the point $(\lambda_1, \lambda_2, 7)$ from the line L_1 is :

रेखा L_1 , बिन्दु $(1, 2, 3)$ से होकर जाती है और z -अक्ष के समांतर है। रेखा L_2 , बिन्दु $(\lambda, 5, 6)$ से होकर जाती है और y -अक्ष के समांतर है। माना $\lambda = \lambda_1, \lambda_2, \lambda_2 < \lambda_1$, के लिए इन दो रेखाओं के बीच न्यूनतम दूरी 3 है। तो बिन्दु $(\lambda_1, \lambda_2, 7)$ की रेखा L_1 से दूरी का वर्ग है :

- (1) 32 (2) 40 (3) 37 (4) 25

Ans. D

Sol. $L_1 : \frac{x-1}{0} = \frac{y-2}{0} = \frac{z-3}{1}$

$$L_2 : \frac{x-\lambda}{0} = \frac{y-5}{1} = \frac{z-6}{0}$$

$$S.D. = \left| \frac{\overline{AB}(\overline{p} \times \overline{q})}{(\overline{p} \times \overline{q})} \right| = |\lambda - 1| = 3$$

$$\lambda = 4 \text{ or } \lambda = -2$$

Let foot of perpendicular from

$$P(4, -2, 7) \text{ is } Q(1, 2, t+3)$$

$$(3, -4, 4-t) \cdot (0, 0, 1) = 0$$

$$t = 4$$

$$PQ^2 = 25$$

11. Let a line passing through the point $(4, 1, 0)$ intersect the line $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ at the point $A(\alpha, \beta, \gamma)$ and

the line $L_2 : x - 6 = y = -z + 4$ at the point $B(a, b, c)$. Then $\begin{vmatrix} 1 & 0 & 1 \\ \alpha & \beta & \gamma \\ a & b & c \end{vmatrix}$ is equal to :

माना बिन्दु $(4, 1, 0)$ से होकर जाने वाली एक रेखा, रेखा $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ को बिन्दु $A(\alpha, \beta, \gamma)$ और रेखा $L_2 : x - 6 = y = -z + 4$

$= y = -z + 4$ को बिन्दु $B(a, b, c)$ पर काटती है। तो $\begin{vmatrix} 1 & 0 & 1 \\ \alpha & \beta & \gamma \\ a & b & c \end{vmatrix}$ बराबर है :

(1) 6

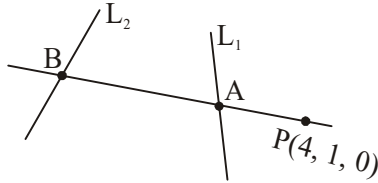
(2) 8

(3) 16

(4) 12

Ans. B

Sol.



$$L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} = p$$

$$A(2p+1, 3p+2, 4p+3)$$

$$L_2 : \frac{x-6}{1} = \frac{y}{1} = \frac{z-4}{-1} = q$$

$$B(q+6, q, 4-q)$$

$$\text{D.R. of PA} = (2p-3, 3p+1, 4p+3)$$

$$\text{D.R. of PB} = (q+2, q-1, 4-q)$$

$$\frac{2p-3}{q+2} = \frac{3p+1}{q-1} = \frac{4p+3}{4-q}$$

$$2pq - 2p - 3q + 3 = 3pq + 6p + q + z$$

$$pq + rp + 4q - 1 = 0 \quad \dots\dots\dots(1)$$

$$12p - 3pq + 4 - q = 4pq + 3q - 4p - 3$$

$$7pq - 16p + 4q - 7 = 0 \quad \dots\dots\dots(2)$$

$$pq = -3$$

$$p = -1$$

$$q = 3$$

$$A(-1, -1, -1)$$

$$B(9, 3, 1)$$

$$\begin{vmatrix} 1 & 0 & 1 \\ -1 & -1 & -1 \\ 9 & 3 & 1 \end{vmatrix} = 8$$

12. Each of angles β and γ that a given line makes the positive y- and z-axes, respectively, is half of the angle that this line makes with the positive x-axes. Then the sum of all possible values of the angle β is :

दी गई एक रेखा के धनात्मक y- और z-अक्षों से क्रमशः बनाए गए कोणों β और γ में प्रत्येक, इस रेखा के धनात्मक x-अक्ष से बनाए गए कोण का आधा है। तो β के सभी संभव मानों का योग है :

(1) $\frac{3\pi}{4}$

(2) π

(3) $\frac{3\pi}{2}$

(4) $\frac{\pi}{2}$

Ans. A

Sol. $\beta = \frac{\alpha}{2}, \gamma = \frac{\alpha}{2}$
 $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

$$\cos^2 \alpha + 2 \cos^2 \frac{\alpha}{2} = 1$$

$$\cos^2 \alpha + \cos \alpha = 0$$

$$\cos \alpha (\cos \alpha + 1) = 0$$

$$\cos \alpha = 0, -1$$

$$\alpha = \frac{\pi}{2}, \pi$$

$$\text{Now, } \beta = \frac{\alpha}{2} \Rightarrow \frac{\pi}{4}, \frac{\pi}{2}$$

$$\text{So, sum is } \frac{3\pi}{4}$$

13. Let A and B be two distinct points on the line $L: \frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$. Both A and B are at a distance $2\sqrt{17}$ from the foot of perpendicular drawn from the point (1, 2, 3) on the line L. If O is the origin, then $\overrightarrow{OA} \cdot \overrightarrow{OB}$ is equal to :

माना रेखा $L: \frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$ पर दो भिन्न बिन्दु A और B हैं। बिन्दु (1, 2, 3) से रेखा L पर डाले गए लंब के पाद से A और

B दोनों $2\sqrt{17}$ की दूरी पर हैं। यदि O मूलबिन्दु है, तो $\overrightarrow{OA} \cdot \overrightarrow{OB}$ बराबर है :

(1) 62

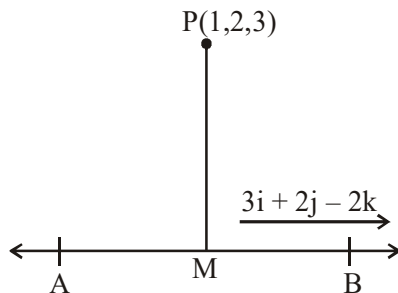
(2) 21

(3) 49

(4) 47

Ans. D

Sol. $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2} = \lambda$



Let M be foot of perpendicular

$$M \{3\lambda + 6, 2\lambda + 7, -2\lambda + 7\}$$

$$\overrightarrow{MP} = \{3\lambda + 5, 2\lambda + 5, -2\lambda + 4\}$$

$$\vec{b} = 3i + 2j - 2k \text{ (Direction ratio of line)}$$

$$\vec{MP} \cdot \vec{b} = 0$$

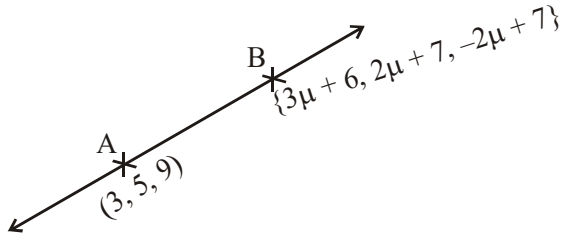
$$(3\lambda + 5) \cdot 3 + (2\lambda + 5) \cdot 2 + (-2\lambda + 4) \cdot (-2) = 0$$

$$9\lambda + 15 + 4\lambda + 10 + 4\lambda - 8 = 0$$

$$17\lambda = -17$$

$$\lambda = -1$$

$$\therefore M \{3, 5, 9\}$$



A.T.Q.

$$|AB| = 2\sqrt{17}$$

$$\sqrt{(3\mu + 3)^2 + (2\mu + 2)^2 + (-2\mu - 2)^2} = \sqrt{17 \times 4}$$

$$9(\mu + 1)^2 + 4(\mu + 1)^2 + 4(\mu + 1)^2 = 17 \times 4$$

$$17(\mu + 1)^2 = 17 \times 4$$

$$\mu + 1 = \pm 2$$

$$\mu = -1 \pm 2$$

$$\mu = 1, -3$$

$$\therefore B \{9, 9, 5\} \text{ or } \{-3, 1, 13\}$$

$$\vec{OA} \cdot \vec{OB} = 9 \times (-3) + 9 \times 1 + 5 \times 13$$

$$= -27 + 9 + 65$$

$$= 47$$

14. Let A be the point of intersection of the lines $L_1 : \frac{x-7}{1} = \frac{y-5}{0} = \frac{z-3}{-1}$ and $L_2 : \frac{x-1}{3} = \frac{y+3}{4} = \frac{z+7}{5}$. Let B and

C be the points on the lines L_1 and L_2 respectively such that $AB = AC = \sqrt{15}$. Then the square of the area of the triangle ABC is :

माना रेखाओं $L_1 : \frac{x-7}{1} = \frac{y-5}{0} = \frac{z-3}{-1}$ और $L_2 : \frac{x-1}{3} = \frac{y+3}{4} = \frac{z+7}{5}$ का प्रतिच्छेदन बिन्दु A है। माना रेखाओं L_1 और L_2

पर क्रमशः बिन्दु B और C इस प्रकार हैं कि $AB = AC = \sqrt{15}$ है। तो त्रिभुज ABC के क्षेत्रफल का वर्ग है :

(1) 57

(2) 63

(3) 60

(4) 54

Ans. D

Sol. $\text{Area} = \frac{1}{2}ab \sin \theta$

$$\Rightarrow \cos \theta = \left| \frac{-2}{(\sqrt{2})\sqrt{50}} \right| = \frac{1}{5}$$

$$\sin \theta = \frac{\sqrt{24}}{5}$$

$$\text{Area} = \frac{1}{2}(\sqrt{15})(\sqrt{15})\left(\frac{\sqrt{24}}{5}\right) = \frac{3}{2}\sqrt{24}$$

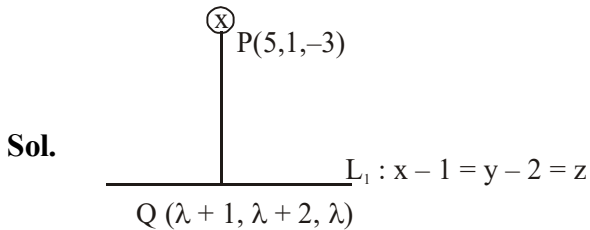
$$(\text{Area})^2 = 54$$

15. Consider the lines $L_1 : x - 1 = y - 2 = z$ and $L_2 : x - 2 = y = z - 1$. Let the feet of the perpendiculars from the point $P(5, 1, -3)$ on the lines L_1 and L_2 be Q and R respectively. If the area of the triangle PQR is A , then $4A^2$ is equal to :

रेखाओं $L_1 : x - 1 = y - 2 = z$ और $L_2 : x - 2 = y = z - 1$ का विचार कीजिए। माना बिन्दु $P(5, 1, -3)$ से रेखाओं L_1 और L_2 पर डाले गए लम्ब के पाद क्रमशः Q और R हैं। यदि त्रिभुज PQR का क्षेत्रफल A है, तो $4A^2$ बराबर है :

- (1) 151 (2) 143 (3) 139 (4) 147

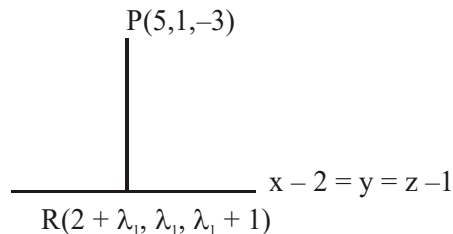
Ans. D



$$\overrightarrow{PQ} \cdot (1, 1, 1) = 0 \Rightarrow (\lambda - 4) + (\lambda + 1) + (\lambda + 3) = 0$$

$$\Rightarrow 3\lambda = 0 \Rightarrow \lambda = 0$$

So, point Q is $(1, 2, 0)$

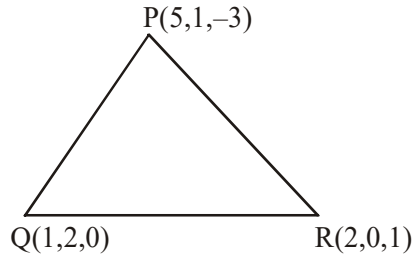


$$\overrightarrow{PR} \cdot (1, 1, 1) = 0$$

$$\Rightarrow (\lambda_1 - 3, \lambda_1 - 1, \lambda_1 + 4) \cdot (1, 1, 1) = 0$$

$$\Rightarrow 3\lambda_1 = 0 \Rightarrow \lambda_1 = 0$$

So, point R is (2, 0, 1)



$$\overrightarrow{PQ} = (-4, 1, 3)$$

$$\overrightarrow{PR} = (-3, -1, 4)$$

$$\overrightarrow{PQ} \times \overrightarrow{PR} = (7, 7, 7)$$

$$\text{so, area (A)} = \frac{1}{2} |\overrightarrow{PQ} \times \overrightarrow{PR}| = \frac{7\sqrt{3}}{2}$$

$$4A^2 = \frac{4 \times 49 \times 3}{4} = 147$$

16. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{3}$. If $\lambda\vec{a} + 2\vec{b}$ and $3\vec{a} - \lambda\vec{b}$ are perpendicular to each other, then the number of values of λ in $[-1, 3]$ is :

माना कि $\vec{a}$ और $\vec{b}$ दो इकाई सदिश इस प्रकार हैं कि उनके बीच का कोण $\frac{\pi}{3}$ है। यदि $\lambda\vec{a} + 2\vec{b}$ और $3\vec{a} - \lambda\vec{b}$ एक दूसरे के लम्बवत् हैं, तो $[-1, 3]$ में λ के मानों की संख्या है :

- (1) 1 (2) 2 (3) 3 (4) 0

Ans. D

Sol. $|\vec{a}| = 1$ $|\vec{b}| = 1$

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \frac{\pi}{3} = \frac{1}{2}$$

$$\text{So } (\lambda\vec{a} + 2\vec{b}) \cdot (3\vec{a} - \lambda\vec{b}) = 0$$

$$3\lambda - \lambda^2 + 6\vec{a} \cdot \vec{b} - \lambda\vec{a} \cdot \vec{b} = 0$$

$$3\lambda - \lambda^2 + \frac{6}{2} - \frac{\lambda}{2} = 0$$

$$2\lambda^2 - 5\lambda + 6 = 0$$

$$D = 25 - 48 < 0 \quad \text{No real value of } \lambda$$

17. Let a line pass through two distinct points P(-2, -1, 3) and Q, and be parallel to the vector $3\hat{i} + 2\hat{j} + 2\hat{k}$. If the

distance of the point Q from the point R(1, 3, 3) is 5, then the square of the area of ΔPQR is equal to :

माना कि एक रेखा दो भिन्न बिन्दुओं P(-2, -1, 3) और Q से होकर गुजरती है और यह सदिश $3\hat{i} + 2\hat{j} + 2\hat{k}$ के समान्तर है। यदि बिन्दु Q की बिन्दु R(1, 3, 3) से दूरी 5 है, तो ΔPQR के क्षेत्रफल का वर्ग बराबर है :

- (1) 148 (2) 140 (3) 136 (4) 144

Ans. C

Sol. P(-2, -1, 3)

equation of line PQ $\vec{r} = (-2, -1, 3) + \lambda(3, 2, 2)$

Let $Q(-2 + 3\lambda, -1 + 2\lambda, 3 + 2\lambda)$
R(1, 3, 3)

$$QR = 5 = \sqrt{(-3 + 3\lambda)^2 + (-4 + 2\lambda)^2 + (2\lambda)^2}$$

$$25 = 17\lambda^2 - 34\lambda + 25$$

$$17\lambda^2 - 34\lambda = 0$$

$$\lambda = 0 \text{ or } \lambda = 2$$

$$Q(4, 3, 7)$$

$$\text{Now Area of } \Delta PQR = \frac{1}{2} |\overrightarrow{PQ} \times \overrightarrow{QR}|$$

$$\overrightarrow{PQ} = 6\hat{i} + 4\hat{j} + 4\hat{k}$$

$$\overrightarrow{QR} = -3\hat{i} + 0\hat{j} - 4\hat{k}$$

$$\overrightarrow{PQ} \times \overrightarrow{QR} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 6 & 4 & 4 \\ -3 & 0 & -4 \end{vmatrix} = -16\hat{i} + 12\hat{j} + 12\hat{k}$$

$$\Delta = \frac{1}{2} \sqrt{256 + 144 + 144} = \frac{\sqrt{544}}{2}$$

$$\Delta^2 = \frac{544}{4} = 136$$

18. Let the position vectors of the vertices A, B and C of a tetrahedron ABCD be $\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + 3\hat{j} - 2\hat{k}$ and $2\hat{i} + \hat{j} - \hat{k}$ respectively. The altitude from the vertex D to the opposite face ABC meets the median line segment through A of the triangle ABC at the point E. If the length of AD is $\frac{\sqrt{110}}{3}$ and the volume of the tetrahedron is $\frac{\sqrt{805}}{6\sqrt{2}}$, then the position vector of E is :

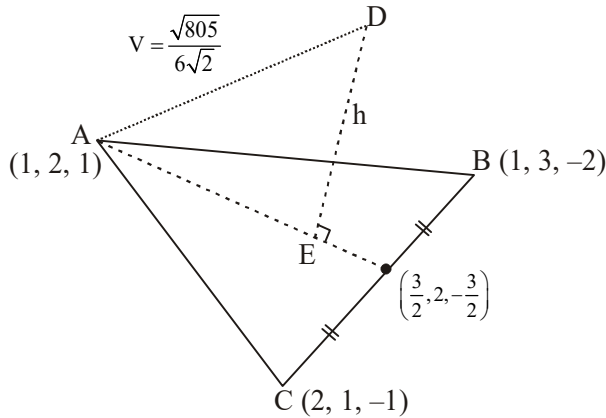
माना कि एक चतुष्फलक ABCD के शीर्षों A, B और C के स्थिति सदिश क्रमशः $\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + 3\hat{j} - 2\hat{k}$ और $2\hat{i} + \hat{j} - \hat{k}$ हैं। शीर्ष D से सम्मुख फलक ABC पर शीर्षलम्ब त्रिभुज ABC के माध्यिका रेखाखण्ड से, A के माध्यम से, बिन्दु E पर मिलता है। यदि AD की

लम्बाई $\frac{\sqrt{110}}{3}$ है और चतुष्फलक का आयतन $\frac{\sqrt{805}}{6\sqrt{2}}$ है, कतो E का स्थिति सदिश है :

- (1) $\frac{1}{6}(12\hat{i}+12\hat{j}+\hat{k})$ (2) $\frac{1}{2}(\hat{i}+4\hat{j}+7\hat{k})$ (3) $\frac{1}{6}(7\hat{i}+12\hat{j}+\hat{k})$ (4) $\frac{1}{12}(7\hat{i}+4\hat{j}+3\hat{k})$

Ans. C

Sol.



Area of $\triangle ABC$

$$\overline{AB} = \hat{j} - 3\hat{k}$$

$$\overline{AC} = \hat{i} - \hat{j} - 2\hat{k}$$

$$\overline{AB} \times \overline{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & -3 \\ 1 & -1 & -2 \end{vmatrix}$$

$$= -5\hat{i} - 3\hat{j} - \hat{k}$$

$$\text{Area of } \triangle ABC = \frac{1}{2}\sqrt{35}$$

$$\text{Area of tetrahedron} = \frac{1}{3}\left(\frac{1}{2}\sqrt{35}\right)h = \frac{\sqrt{805}}{6\sqrt{2}}$$

$$h = \sqrt{\frac{23}{2}}$$

$$AE = \sqrt{\frac{110}{9} - \frac{23}{2}} = \sqrt{\frac{220 - 207}{18}} = \sqrt{\frac{13}{18}}$$

Equation of AE

$$\text{d.r. of AE} = \left(\frac{1}{2}, 0, -\frac{5}{2}\right) = (1, 0, -5)$$

$$\frac{x-1}{1} = \frac{y-2}{0} = \frac{z-1}{-5} = \lambda$$

$$\text{Let } E(\lambda + 1, 2, -5\lambda + 1)$$

$$A(1, 2, 1)$$

$$AE = \sqrt{\frac{13}{18}}$$

$$\lambda^2 + 25\lambda^2 = \frac{13}{18}$$

$$E\left(\frac{7}{6}, 2, \frac{1}{6}\right)$$

$$26\lambda^2 = \frac{13}{18}$$

$$\lambda = \pm \frac{1}{6}$$

19. Let the arc AC of a circle subtend a right angle at the centre O. If the point B on the arc AC, divides the arc AC such that $\frac{\text{length of arc AB}}{\text{length of arc BC}} = \frac{1}{5}$, and $\overrightarrow{OC} = \alpha \overrightarrow{OA} + \beta \overrightarrow{OB}$, then $\alpha + \sqrt{2}(\sqrt{3}-1)\beta$ is equal to :

माना कि एक वृत्त का चाप AC, उसके केन्द्र O पर एक समकोण अंतरित करता है। यदि चाप AC पर बिन्दु B, चाप AC को इस प्रकार विभाजित करता है कि $\frac{\text{चाप AB की लम्बाई}}{\text{चाप BC की लम्बाई}} = \frac{1}{5}$ और $\overrightarrow{OC} = \alpha \overrightarrow{OA} + \beta \overrightarrow{OB}$ है, तो $\alpha + \sqrt{2}(\sqrt{3}-1)\beta$ बराबर है :

$$(1) 2 + \sqrt{3}$$

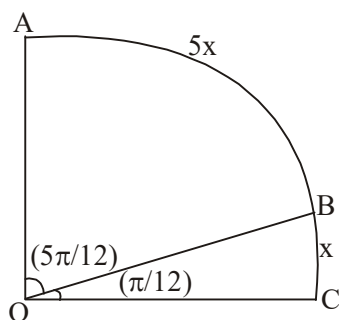
$$(2) 2 - \sqrt{3}$$

$$(3) 5\sqrt{3}$$

$$(4) 2\sqrt{3}$$

Ans. B

Sol.



$$OC = OA = OB = r$$

$$6x = \frac{\pi}{2}$$

$$x = \frac{\pi}{12}$$

$$OC = OA = OB = r$$

$$\overrightarrow{OC} = \alpha \overrightarrow{OA} + \beta \overrightarrow{OB}$$

$$\text{dot with } \overrightarrow{OA}$$

$$0 = \alpha |\overrightarrow{OA}|^2 + \beta \overrightarrow{OA} \cdot \overrightarrow{OB}$$

$$0 = r^2 \alpha + r^2 \beta \cos \frac{\pi}{12}$$

$$\alpha + \frac{\sqrt{3}+1}{2\sqrt{2}} \beta = 0 \quad \dots(1)$$

$$\overrightarrow{OC} = \alpha \overrightarrow{OA} + \beta \overrightarrow{OB}$$

dot with $\overrightarrow{OC}$

$$r^2 = \alpha r^2 \cos \frac{\pi}{12} + \beta r^2 \cos \frac{5\pi}{12}$$

$$\beta \frac{\sqrt{3}-1}{2\sqrt{2}} = 1 \quad (\sqrt{3}-1)\beta = 2\sqrt{2}$$

$$\beta = \sqrt{2}(\sqrt{3}+1)$$

$$\alpha + \frac{\sqrt{2}(\sqrt{3}+1)^2}{2\sqrt{2}} = 0 \quad \alpha + \sqrt{2}(\sqrt{3}-1)\beta$$

$$\alpha = -\frac{3+1+2\sqrt{3}}{2} = -2-\sqrt{3}+4$$

$$\alpha = -2-\sqrt{3} = 2-\sqrt{3}$$

20. Let the point A divide the line segment joining the points P(-1, -1, 2) and Q(5, 5, 10) internally in the ratio

$r : 1 (r > 0)$. If O is the origin and $(\overrightarrow{OQ} \cdot \overrightarrow{OA}) - \frac{1}{5} |\overrightarrow{OP} \times \overrightarrow{OA}|^2 = 10$, then the value of r is :

माना कि बिन्दु A, बिन्दुओं P(-1, -1, 2) और Q(5, 5, 10) को मिलाने वाले रेखाखण्ड को अन्तः $r : 1 (r > 0)$ के अनुपात में विभाजित

करता है। यदि O मूल बिन्दु है और $(\overrightarrow{OQ} \cdot \overrightarrow{OA}) - \frac{1}{5} |\overrightarrow{OP} \times \overrightarrow{OA}|^2 = 10$ है, तो r का मान है :

- (1) 3 (2) 7 (3) $\sqrt{7}$ (4) 14

Ans. B

Sol. $A \left[\frac{5r-1}{r+1}, \frac{5r-1}{r+1}, \frac{10r+2}{r+1} \right]$

$$(\overrightarrow{OQ} \cdot \overrightarrow{OA}) - \frac{|\overrightarrow{OP} \times \overrightarrow{OA}|^2}{5} = 10 \quad \dots(1)$$

$$(\overrightarrow{OQ} \cdot \overrightarrow{OA}) = \frac{5}{r+1} (30r+2)$$

$$|\overrightarrow{OP} \times \overrightarrow{OA}|^2 = \frac{r^2}{(r+1)^2} (800)$$

So, by equation (1)

$$\frac{10}{r+1} (15r+1) - \frac{r^2 (800)}{5(r+1)^2} = 10$$

$$2r^2 - 14r = 0$$

$$r = 7, r \neq 0$$